Hypothesis Testing

This is performed under inferential stats.

Z - Test

When To Use T-test Vs Z -Test

Do you know the std dev. (σ) of Population

- Y → Is the sample size above 30?
  - Y → Use Z-test
  - N → Use T-test
- N → Use T-test

Z - Test

eg The average heights of all residents in a city is 168cm with a population std $\sigma = 3.9$. A doctor believe the mean to be different. He measured a height of 36 individuals & found the avg to be 169.5cm.

1) State Null & Alternate Hypothesis
2) At a 95% confidence Interval, is there enough evidence to reject the null hypothesis

Ans $\mu = 168 cm$, $\sigma = 3.9$, $n = 36$

$\bar{x} = 169.5 cm$

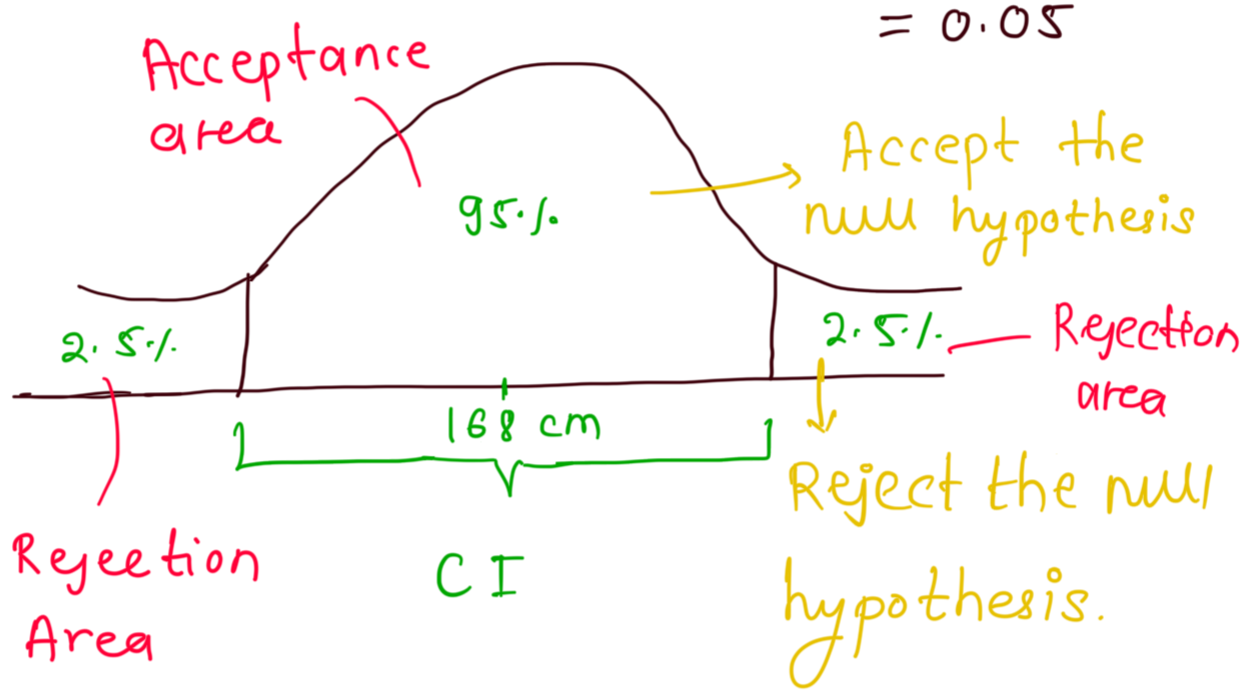
a) Null hypothesis $H_0 \Rightarrow \mu = 168 cm$

b) Alternate hypothesis $H_1 \Rightarrow \mu \neq 168cm$

This is called 2 tail test

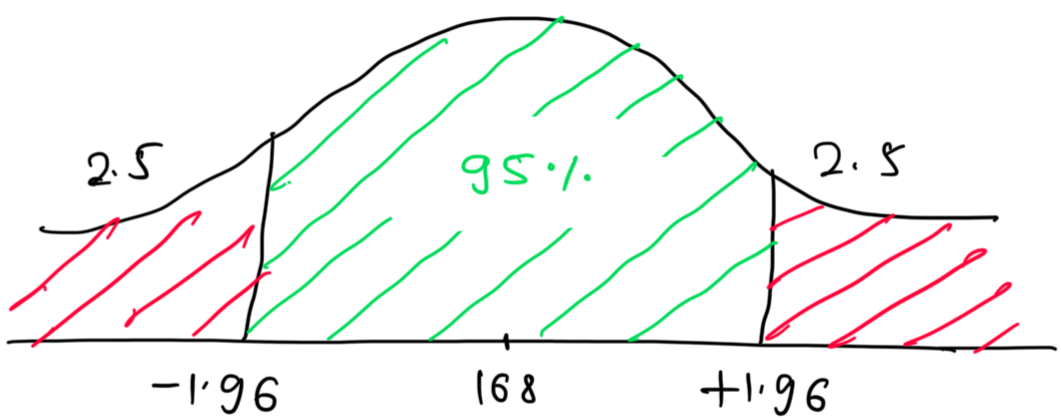
c) $CI = 0.95 \Rightarrow 95\%$, $\alpha = 1 - CI = 1 - 0.95$

$= 0.05$



Area under the curve = 1

$\therefore 1 - 0.025 = 0.9750$ $0.025 \Rightarrow 2.5\%$



1.96 comes from Z-table check in table for value 0.9750 then it gave 1.96 which is decision boundary

If Z-value falls between -1.96 to +1.96 then we fail to reject the null hypothesis i.e we accept the null hypothesis

d) statistical Analysis

$$Z\text{-test} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

n = sample size

$\sigma/\sqrt{n}$ = std error

$$Z\text{-score} = \frac{\bar{x} - \mu}{\sigma}$$

$$= \frac{169.5 - 168}{3.9/\sqrt{36}} = 2.31$$

If z-test is less than −1.96 or greater than +1.96, reject the null hypothesis.

$2.31 > 1.96$ ∴ We reject the null hypothesis.

Significance Level (α)

It is a predefined threshold (commonly 0.05 or 0.01) representing the maximum acceptable risk of rejecting a true null hypothesis, known as a Type 1 error (false positive). It determines the probability of concluding a result is statistically significant when it is actually due to chance

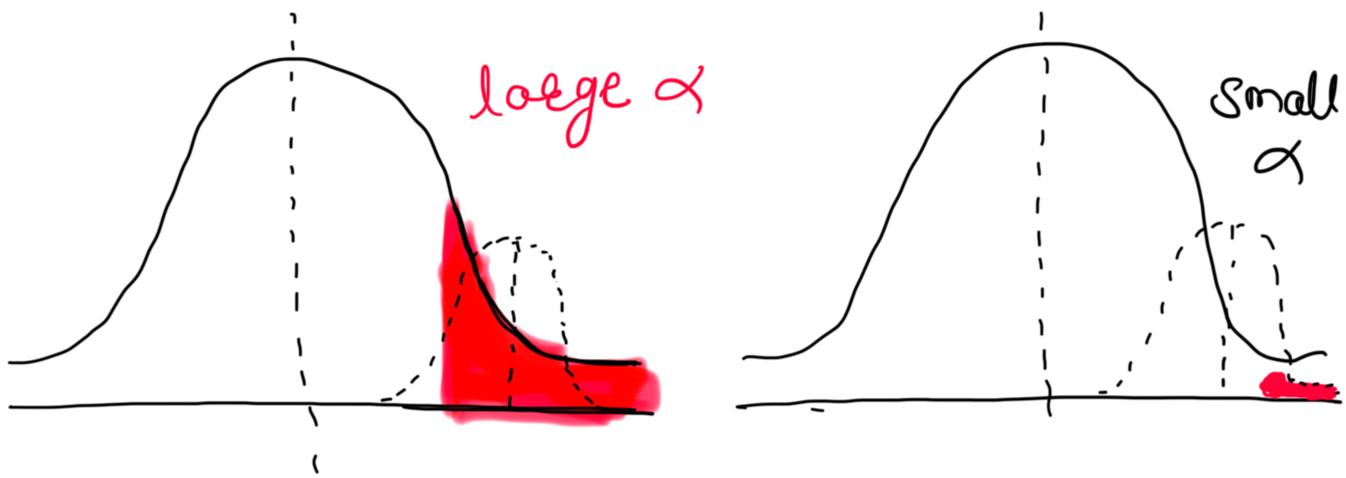
In short

The probability that we make the

error to reject the null hypothesis (H_0) when the null hypothesis is true.

small α : less likely to incorrectly reject H_0

large α : more likely to incorrectly reject H_0



How to check whether it's one tail or two tail

Case 1 - Two tailed test

If H_1 talks about "difference" or "not equal to"

$$H_1 : \mu \neq \mu_0$$

Case II – One-tailed Test

If H_1 talks about direction (only one side)

Right tailed $H_1 : \mu > \mu_0$

Left tailed $H_1 : \mu < \mu_0$

Is there any difference? → Two-tailed

Has it increased / decreased? → One-tailed

μ = True Population Mean

μ_0 = Assumed / claimed mean

$\bar{x}$ = Sample mean (from data)

2) A factory manufactures bulbs with an avg waranty of 5 years with std deviation of 0.50. A worker believes that the bulb will malfunction in less than 5 years. He tests a sample of 40 bulbs & finds the avg

time to be 4.8 years.

a) State null & alternate hypothesis

b) At a 2% significance level, is there enough evidence to support idea that the warranty should be revised

Sol

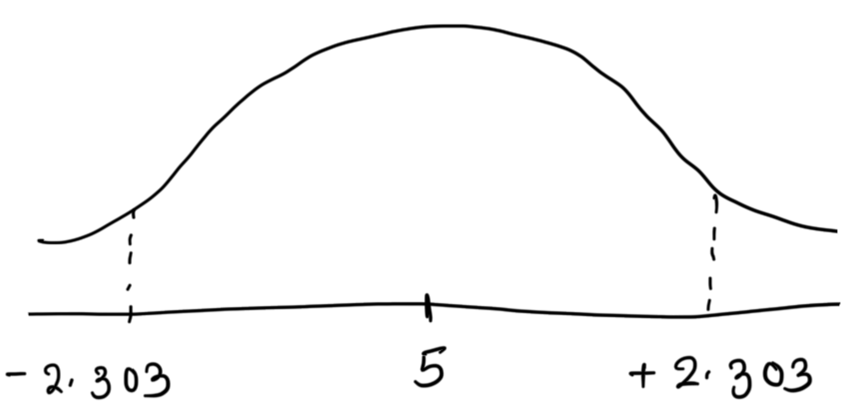
1) Null hypothesis H_0 - years = 5 yrs

2) Alternate hypothesis H_1 - years $\neq$ 5 yrs

3) $\alpha = 2\% = 0.02$

$\therefore$ area under the curve $= 1 - 0.01$

$= 0.99$



$$Z\text{-test} = \frac{\bar{x} - \mu}{} = \frac{4.8 - 5}{} = \frac{-0.2}{}$$

$\sigma/\sqrt{n}$ $\quad 0.5/\sqrt{40}$ $\quad 0.07906$

$= -2.53$

$-2.303 > 2.53$

We failed to accept null hypothesis

i.e We reject the null hypothesis

OR

1) Hypothesis

Worker believes that bulbs fail in less than 5 years, so its a left tail test

$H_0 : \mu = 5$

$H_1 : \mu < 5$

2) Test at 2% significance

a) $Z = -2.53$

b) Critical value (left tail, $\alpha = 0.02$)

$$Z_{0.02} \approx -2.054$$

c) Decision

- Our test statistics : $Z = -2.53$
- Critical cutoff : -2.054

Since

$$-2.53 < -2.054$$

We reject H_0